Session 9B: 2D Geometry — Shapes, Symmetry, and Space

Phase 2 — Classical Techniques | 75 min

How functions create shapes. No calculus — pure algebra and geometric reasoning.

Part A: Composition and Inverse — The Geometry of Reversing

Example 1: Composition as a Two-Stage Pipeline

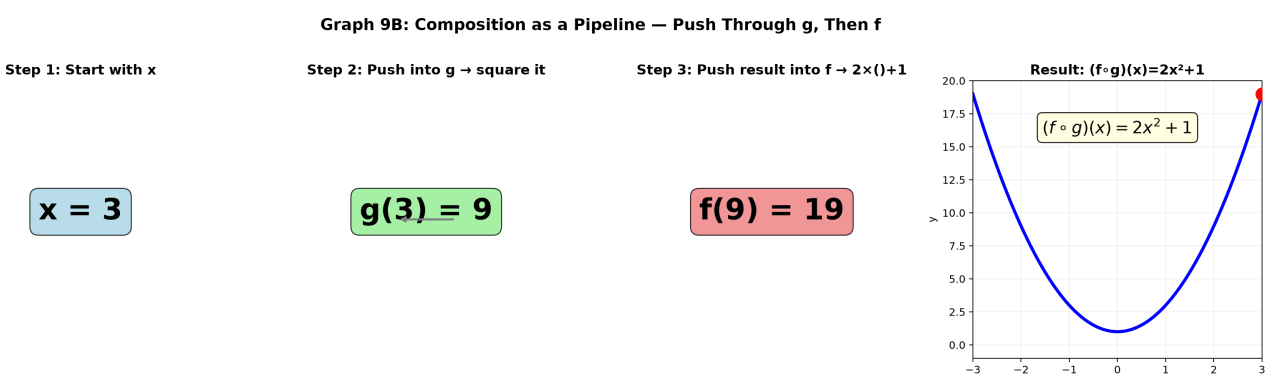
$f(x) = 2x + 1$, $g(x) = x^2$.

$(f \circ g)(x) = f(g(x))$: ① Square it → x^2. ② Double and add 1 → $2x^2 + 1$.

$(g \circ f)(x) = g(f(x))$: ① Double and add 1 → $2x + 1$. ② Square → $(2x + 1)^2 = 4x^2 + 4x + 1$.

Different results! Composition is not commutative: $f \circ g \neq g \circ f$ in general.

Geometric view: g warps the x-values, then f warps the output. Each function is a transformation of the number line.



Graph 9B-S1: Composition as a pipeline — x=3 enters, g squares it to 9, f doubles+1 to get 19. Final formula: (f∘g)(x)=2x²+1. Each box is one transformation stage.

Example 2: Composition Shrinks the Domain

$f(x) = \sqrt{x}, g(x) = x - 3.$

$f \circ g$: $\sqrt{x-3}$. Inner g needs no restriction, but f only accepts ≥ 0.
$x - 3 \geq 0 \rightarrow x \geq 3$. Domain shrinks to $[3, \infty)$.

$g \circ f$: $\sqrt{x} - 3$. f first: needs $x \geq 0$. g accepts anything. Domain stays $[0, \infty)$.

Rule: The inner function's range must fit inside the outer function's domain.

Example 3: Inverse — Swap Input and Output, Reflect Across $y = x$

$f(x) = 3x - 6.$

① Write $y = 3x - 6$. ② Solve for x: $x = \frac{y+6}{3}$. ③ Swap x and y: $f^{-1}(x) = \frac{x+6}{3}$.

Geometric meaning: Every point (a, b) on f becomes (b, a) on f^{-1}. The graph of f^{-1} is the reflection of f across the line $y = x$.

Example 4: Rational Inverse

$f(x) = \frac{2x+1}{x-3}.$

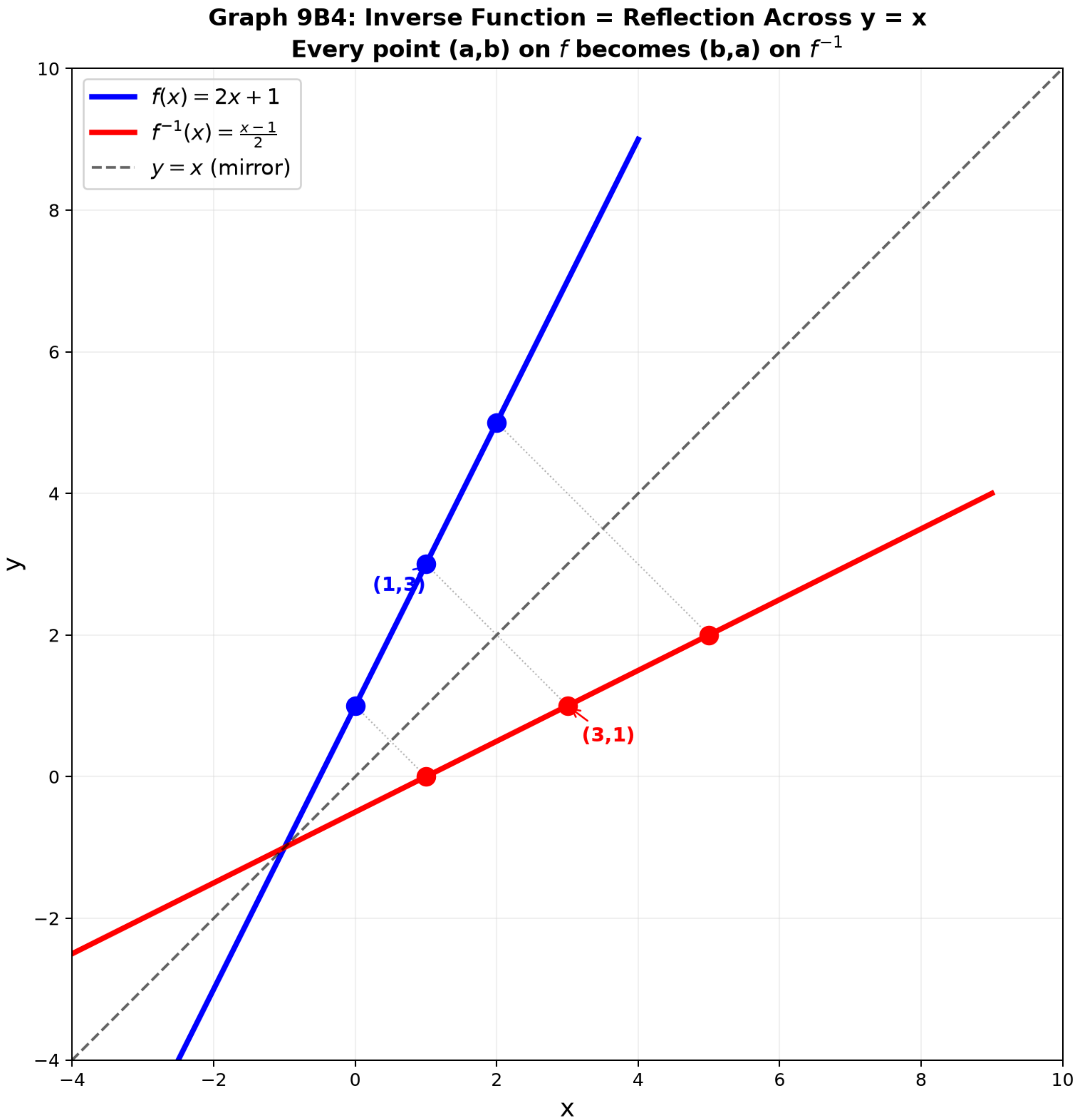
① $y(x-3) = 2x + 1$. ② $yx - 3y = 2x + 1$. ③ $yx - 2x = 3y + 1$. ④ $x(y-2) = 3y + 1$. ⑤ $x = \frac{3y+1}{y-2}$. ⑥ $f^{-1}(x) = \frac{3x+1}{x-2}$.

Example 5: When You Must Snip the Domain

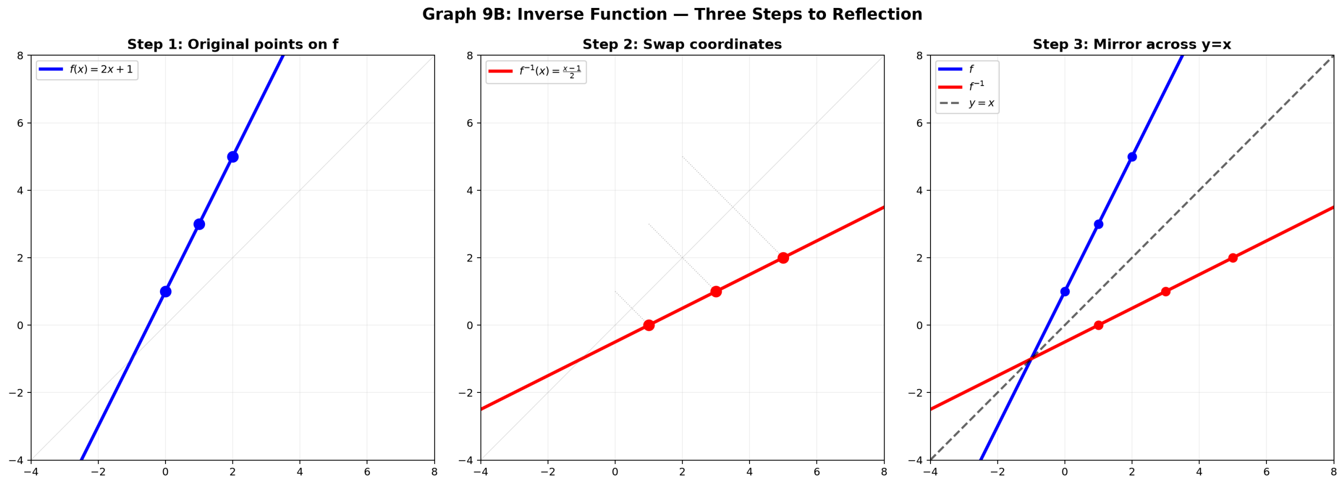
$f(x) = x^2$. Push in $2 \to 4$. Push in $-2 \to 4$. Two inputs → one output. No inverse unless we restrict.

Snip to right half: domain $[0, \infty)$ → inverse $f^{-1}(x) = \sqrt{x}$.
Snip to left half: domain $(-\infty, 0]$ → inverse $f^{-1}(x) = -\sqrt{x}$.

One-to-one test (horizontal line test): If any horizontal line crosses the graph more than once, the function is not invertible until you restrict the domain.



Graph 9B4: f(x)=2x+1 (blue) and its inverse $f^{-1}(x)=(x-1)/2$ (red). Every point (a,b) on f becomes (b,a) on f^{-1}, reflected across the dashed line y=x.



Graph 9B-S2: Building the inverse in three steps. Step 1 — Plot original points on f. Step 2 — Swap (x,y)→(y,x) coordinates. Step 3 — The result is a perfect mirror across y=x (dashed black). Notice (0,1)→(1,0), (1,3)→(3,1), (2,5)→(5,2).

Part B: Symmetry — The Two Mirror Types

Example 6: Even Functions — y-Axis Mirror

$f(-x) = f(x)$ for all x in the domain. The right half determines the left half.

$f(x) = x^2$, $f(x) = \cos x$, $f(x) = |x|$, $f(x) = \frac{1}{x^2+1}$.

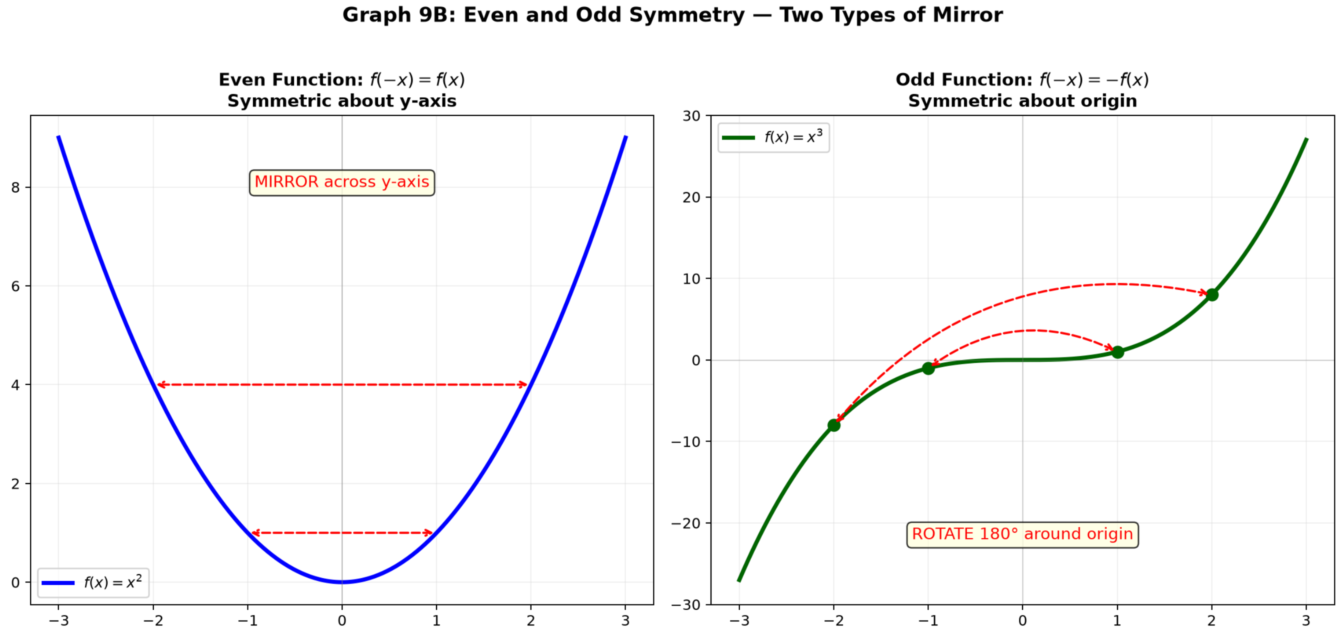
Test: Replace x with $-x$. If the formula doesn't change, it's even.
Drawing trick: Draw only $x \geq 0$, then mirror across the y-axis.

Example 7: Odd Functions — Origin Rotation

$f(-x) = -f(x)$ for all x. The graph spins 180° around the origin onto itself.

$f(x) = x^3$, $f(x) = \sin x$, $f(x) = \frac{1}{x}$, $f(x) = x|x|$.

Test: Replace x with $-x$. If you get $-f(x)$, it's odd.
Drawing trick: Draw only $x \geq 0$, then rotate 180° around the origin for $x < 0$.



Graph 9B: Left — Even function (x^2) mirrors across the y-axis. Right — Odd function (x^3) rotates 180° around the origin. Every function decomposes into even + odd parts.

Part C: Conic Sections

$f(-x) = -f(x)$ for all x. The graph spins 180° around the origin onto itself.

Key fact: Every function decomposes into even + odd:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}.$$

Part C: Conic Sections — The Four Classic Shapes

> Cut a double cone with a plane at different angles. Four shapes emerge.

Example 8: The Circle — Constant Distance from a Center

Equation: $(x-h)^2 + (y-k)^2 = R^2$. Center (h, k), radius R.

$x^2 + y^2 = 25$: center $(0, 0)$, radius 5.

Not a function (fails vertical line test). But we understand it geometrically: all points exactly 5 units from the origin.

General form: $x^2 + y^2 + Dx + Ey + F = 0$. Complete the square in both x and y to find center and radius.

Example 9: The Ellipse — Stretched Circle

Equation: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

a = semi-major axis (horizontal radius). b = semi-minor axis (vertical radius).

$\frac{x^2}{16} + \frac{y^2}{9} = 1$: extends ± 4 horizontally, ± 3 vertically.

Geometric definition: All points where the sum of distances to two fixed foci is constant.

Eccentricity: $e = \sqrt{1 - \frac{b^2}{a^2}}$ $(a \geq b)$. $e = 0$ = circle, $e \to 1$ = very flat.

Example 10: The Parabola — Focus and Directrix

Equation (vertical): $y = \frac{1}{4p}(x-h)^2 + k$. Vertex (h, k). Focus at $(h, k+p)$. Directrix $y = k - p$.

$y = \frac{1}{2}x^2$: $4p = 2 \to p = \frac{1}{2}$. Vertex $(0, 0)$, focus $(0, \frac{1}{2})$, directrix $y = -\frac{1}{2}$.

Geometric definition: Every point on the parabola is equidistant from the focus and the directrix.

Example 11: The Hyperbola — Two Mirrored Branches

Equation: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ (horizontal opening).

Asymptotes: $y - k = \pm\frac{b}{a}(x - h)$. The branches hug these lines as $x \to \pm\infty$.

$\frac{x^2}{4} - \frac{y^2}{9} = 1$: opens left/right. Asymptotes $y = \pm\frac{3}{2}x$.

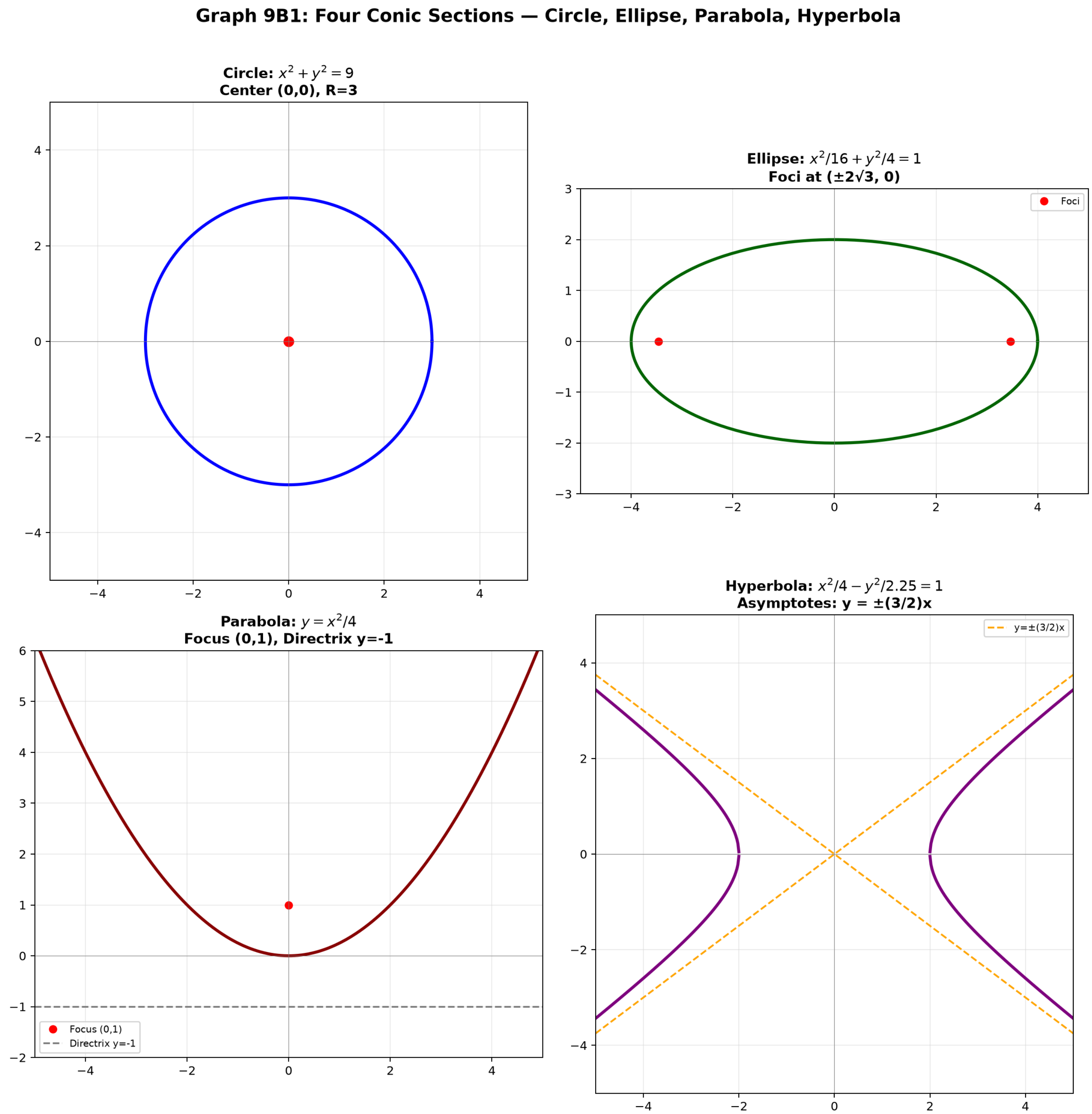
Geometric definition: All points where the absolute difference of distances to two foci is constant.

Example 12: Identifying Conics from General Form

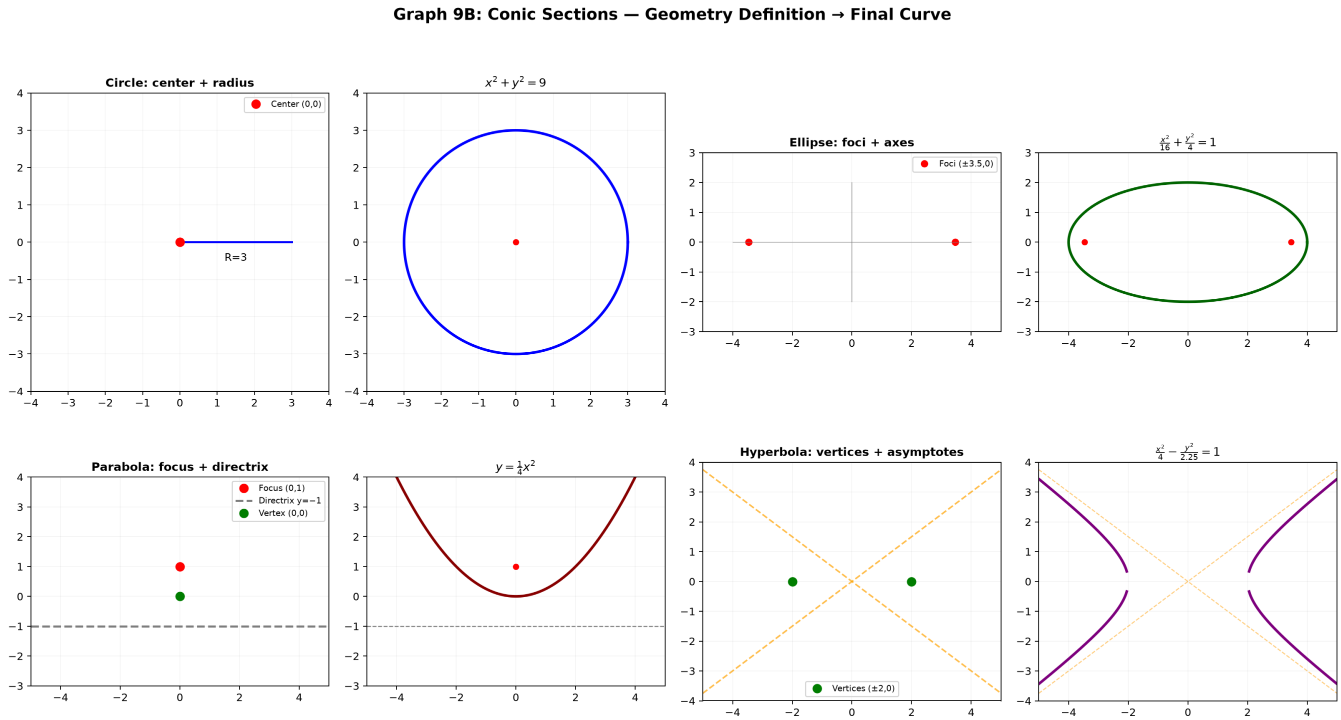
$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$

Discriminant $B^2 - 4AC$	Type
< 0	Ellipse (or circle if $A = C$)
$= 0$	Parabola
> 0	Hyperbola

Example: $x^2 + 4y^2 - 6x + 8y + 9 = 0$. $B = 0, A = 1, C = 4$. $B^2 - 4AC = -16 < 0$ → Ellipse.
Complete squares: $(x - 3)^2 + 4(y + 1)^2 = 4 \to \frac{(x-3)^2}{4} + \frac{(y+1)^2}{1} = 1.$



Graph 9B1: The four conic sections with their key features labeled — foci, directrix, asymptotes. All arise from cutting a double cone at different angles.



Graph 9B-S3: Each conic built in two stages. Left column — geometric definition (center+radius, foci+sum, focus+directrix, vertices+asymptotes). Right column — the resulting curve with its equation. The parabola $y = x^2/4$ opens upward because p=1>0. The hyperbola branches hug the orange asymptotes.

Part D: Parametric Curves — Describing Motion

Example 13: Parametric Circle and Ellipse

Circle: $(x(t), y(t)) = (R\cos t,\ R\sin t), t \in [0, 2\pi]$.

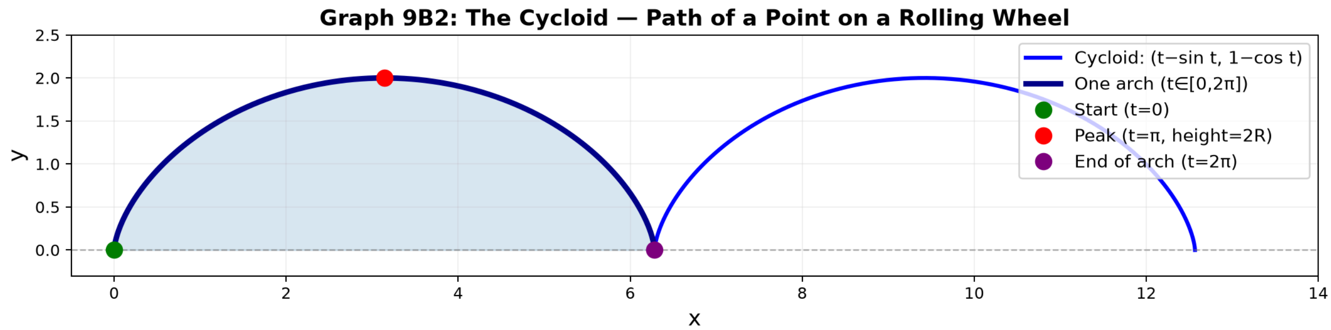
Ellipse: $(x(t), y(t)) = (a\cos t,\ b\sin t), t \in [0, 2\pi]$.

As t runs from 0 to 2π, the point traces the curve counterclockwise. The parameter t is not the angle of the point — it's the angle in the "unwrapped" circle.

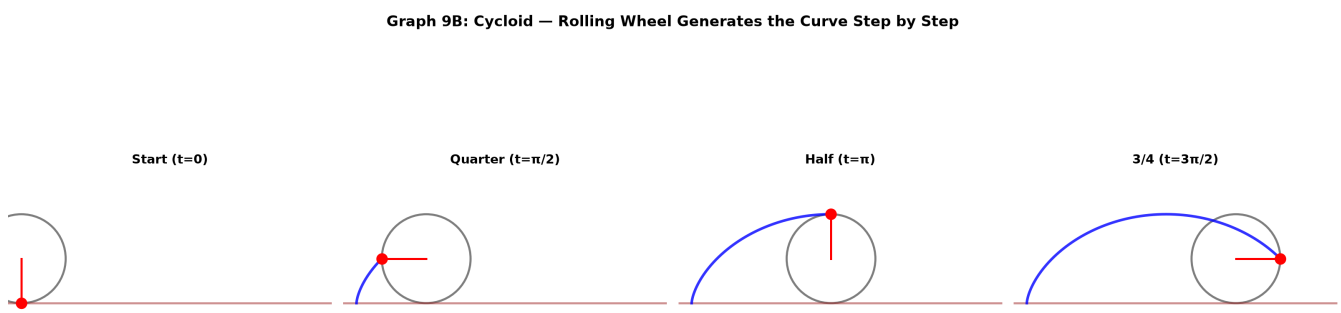
Example 14: The Cycloid — A Wheel's Path

A point on a rolling wheel of radius R traces:
$(x(t), y(t)) = (R(t - \sin t),\ R(1 - \cos t)), t \geq 0.$

One arch: $t \in [0, 2\pi]$. The point starts at $(0, 0)$, rises to height $2R$, and returns to the x-axis at $(2\pi R, 0)$.



Graph 9B2: Two arches of the cycloid. The blue arch (t∈[0,2π]) shows one full rotation. The peak reaches height 2R. The base spans 2πR.



Graph 9B-S4: The cycloid generated step by step. A wheel of radius R=1 rolls rightward. At t=0 the point touches the ground. At t=π/2 it's at height R, at t=π it's at the peak (2R), at t=3π/2 it's descending. The red spoke shows which point on the wheel is being traced. The blue curve is the trail left behind.

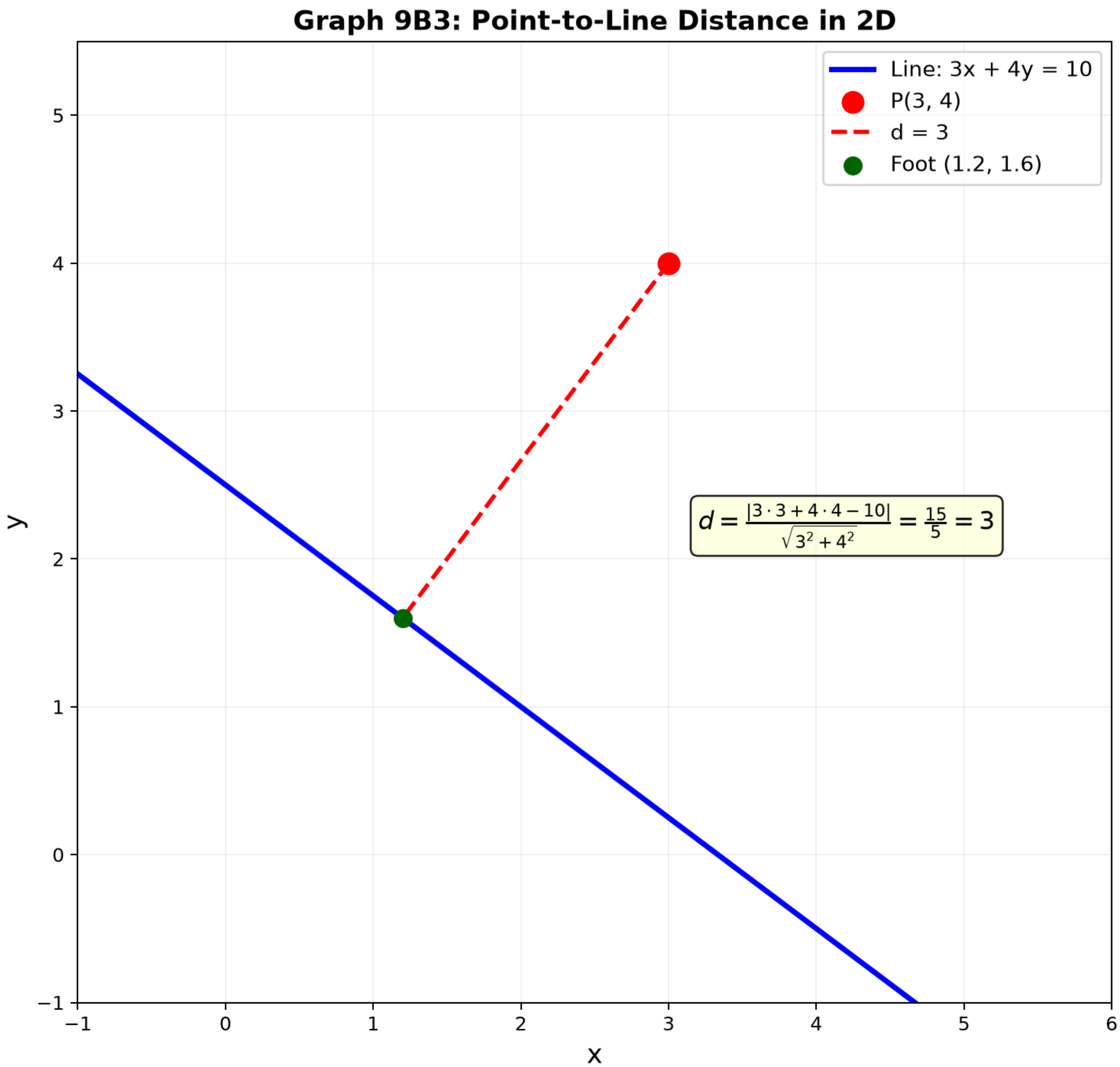
Part E: Distance in 2D — Point, Line, Curve

Example 15: Point-to-Line Distance

Point (x_0, y_0) to line $ax + by + c = 0$:

$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2+b^2}}$.

$(3,4)$ to $3x + 4y - 10 = 0$: $d = \frac{|9+16-10|}{5} = \frac{15}{5} = 3$.



Graph 9B3: The shortest distance from P(3,4) to the line 3x+4y=10 is the perpendicular segment (red dashed). The foot is at (1.2, 1.6). Distance = 3.

Why it works: The numerator is how "wrong" the line equation is at that point. The denominator normalizes for the line's steepness.

Example 16: Distance Between Two Curves — Shortest Gap

The shortest distance between $y = f(x)$ and $y = g(x)$ is the minimum of $\sqrt{(x_1 - x_2)^2 + (f(x_1) - g(x_2))^2}$.

For simple cases, the shortest segment is perpendicular to both curves.

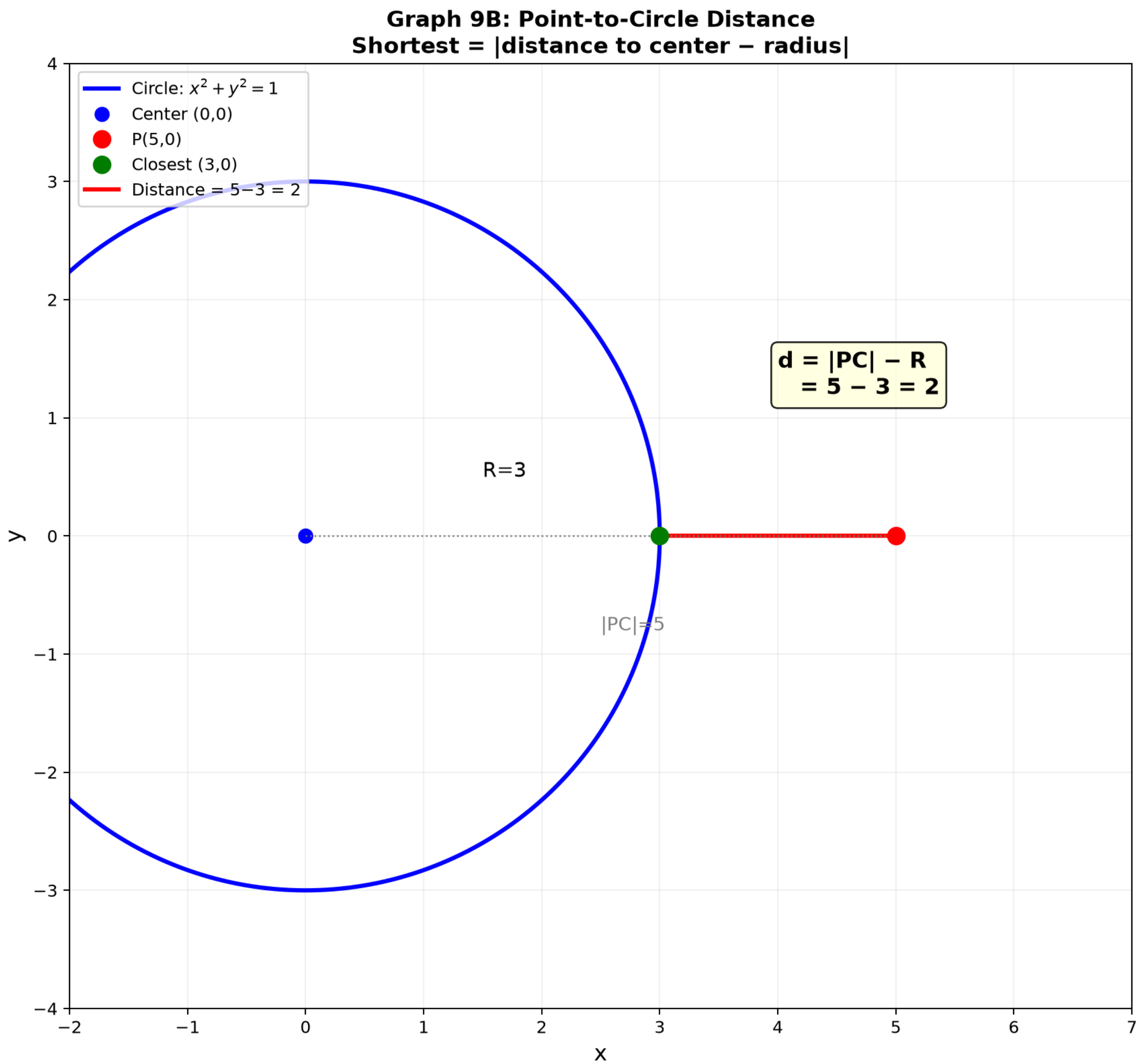
Example: Distance between $y = x^2$ and $y = x - 1$. The line connecting closest points is perpendicular to $y = x - 1$. But without calculus, we can estimate by trying values: at $x = 0.5$, curves at $(0.5, 0.25)$ and $(0.5, -0.5)$, distance 0.75.

Example 17: Point-to-Circle Distance

Point (x_0, y_0) to circle center (h, k), radius R:
Distance = $|\sqrt{(x_0 - h)^2 + (y_0 - k)^2} - R|$.

$(5, 0)$ to circle $x^2 + y^2 = 9$: distance to center is 5, subtract radius $3 \rightarrow 2$.

If the point is inside: subtract the distance from R instead.



Graph 9B: Distance from P(5,0) to circle $x^2+y^2=9$ is |5–3|=2 — the straight line from P through the center to the closest circle point.

> **Up to here**: Composition = pipeline. Inverse = $y = x$ reflection. Even/odd symmetry.
> Conics: circle (constant radius), ellipse (sum to foci), parabola (focus=directrix), hyperbola (difference to foci).
> Parametric curves describe motion. Distance formulas in 2D.

Common Mistakes

Mistake 1: $(f \circ g)^{-1} = f^{-1} \circ g^{-1}$

Wrong. The inverse reverses the order: $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. Socks then shoes → shoes off then socks off.

Mistake 2: Treating a circle as a function

Wrong: "Let $f(x) = \pm\sqrt{25 - x^2}$." **Right**: A circle is not a function of x. Solve implicitly or use parametric form.

Mistake 3: Confusing a and b in ellipse

Wrong: Always $a > b$. **Right**: a is the denominator under $(x - h)^2$ regardless of size. The major axis is the longer one — it could be vertical if $b > a$. Check which denominator is bigger.

What We Just Did

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(1) Composition: inside→outside pipeline. Inverse: reflect across y=x.
    Snip domain if not one-to-one.

(2) Conic sections: circle, ellipse, parabola, hyperbola.
    Each has a geometric definition beyond its equation.

(3) Parametric curves: (x(t), y(t)) describes motion.
    Circle, ellipse, cycloid.

(4) Distance: point-to-line, point-to-circle, between curves.
    Perpendicular = shortest.
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Practice 1

Find $(f \circ g)(x)$ and $(g \circ f)(x)$ for $f(x) = \frac{1}{x}$, $g(x) = x^2 + 1$. State the domain of each composition.

→ Reference: **Example 1, 2**

> Solutions: Solutions

Practice 2

Find the inverse of $f(x) = \frac{3x-1}{x+4}$.

→ Reference: **Example 4**

> Solutions: Solutions

Practice 3

Classify each conic and find its key features:
(a) $x^2 + y^2 - 4x + 6y - 3 = 0$
(b) $4x^2 + 9y^2 = 36$
(c) $y^2 - x^2 = 4$

→ Reference: **Examples 8-12**

> Solutions: Solutions

Practice 4

Find the distance from $(3, 1)$ to the line $4x + 3y = 10$.

→ Reference: **Example 15**

> Solutions: Solutions

Practice 5

A point moves as $(x(t), y(t)) = (3\cos t,\ 2\sin t)$ for $t \in [0, 2\pi]$. What shape does it trace? Where is the point at $t = \pi/3$?

→ Reference: **Example 13**

> Solutions: Solutions

Practice 6: Real Battle

Find the shortest distance from the origin to the line that is tangent to the circle $(x-3)^2 + (y-4)^2 = 1$ at its closest point to the origin. (Hint: the closest point on the circle lies on the line from origin to center.)

→ Reference: **Example 8, 15, 17**

> Solutions: Solutions

Basic Algebra Drill — 2D Geometry (10 Problems)

> Pure computation.

D1. If $f(x) = 3x - 2$ and $g(x) = \sqrt{x}$, find $(f \circ g)(4)$.

D2. Find the inverse of $f(x) = 5x + 2$. Verify that $f(f^{-1}(x)) = x$.

D3. Is $f(x) = x^4 - 3x^2$ even, odd, or neither? Show your test.

D4. Find the center and radius of $x^2 + y^2 + 6x - 10y + 18 = 0$.

D5. Find the vertices and foci of $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

D6. Find the vertex and focus of $y = \frac{1}{8}x^2$.

D7. Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

D8. Parametrize a circle of radius 5 centered at $(2, -3)$.

D9. Find the distance from $(-1, 5)$ to the line $2x - y + 3 = 0$.

D10. A point moves as $(x(t), y(t)) = (2\cos t,\ 2\sin t)$. At what t values is $y = \sqrt{2}$?

> Solutions: Solutions

Advanced Algebra Drill — 2D Geometry (10 Problems)

> Multi-step geometric reasoning.

A1. Show that $(f \circ g)^{-1}(x) = (g^{-1} \circ f^{-1})(x)$ for $f(x) = 2x + 1, g(x) = x^3$.

A2. Decompose $f(x) = e^x + e^{-x}$ into its even and odd parts.

A3. Find the equation of the parabola with focus $(2, 3)$ and directrix $y = -1$.

A4. A hyperbola has asymptotes $y = \pm\frac{2}{3}x$ and passes through $(3, 0)$. Find its equation.

A5. Find the distance between the two parallel lines $3x + 4y - 5 = 0$ and $3x + 4y + 15 = 0$.

A6. Find the point on the circle $x^2 + y^2 = 25$ that is closest to $(7, 1)$.

A7. The parametric curve $(x(t), y(t)) = (t^2, t^3)$ is called a semicubical parabola. At what t does it pass through $(4, 8)$? Eliminate t to find the Cartesian equation.

A8. A line through $(1, 2)$ with slope m intersects the circle $x^2 + y^2 = 5$ at two points. For what values of m is the line tangent (only one intersection)?

A9. Find the equation of the ellipse with foci at $(\pm 3, 0)$ that passes through $(0, 4)$.

A10. Four points form a quadrilateral: $(0, 0)$, $(6, 0)$, $(4, 4)$, $(0, 3)$. Is $(2, 2)$ inside it? Use distance/area reasoning.

> Solutions: Solutions

Today's Procedure

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Step 1: Composition = two-stage transformation. Inverse = swap + reflect across y=x.
        Snip domain if not one-to-one.

Step 2: Conic sections — four shapes from one double cone.
        Circle (R constant), ellipse (sum to foci), parabola (focus=directrix),
        hyperbola (difference to foci). Identify by discriminant B²-4AC.

Step 3: Distance — point-to-line formula. Point-to-circle = |distance to center - R|.
        Parametric curves: (x(t), y(t)) = motion in the plane.
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Terminology

What we call it	Math term	Notation
connecting functions	composition	$(f \circ g)(x)$
swapping input/output	inverse function	$f^{-1}(x)$
y-axis mirror	even function	$f(-x) = f(x)$
origin rotation	odd function	$f(-x) = -f(x)$
one-to-one	injective	horizontal line test
circle/ellipse/parabola/ hyperbola	conic sections	from $Ax^2 + Bxy + Cy^2 + \cdots = 0$

What we call it	Math term	Notation
focus, directrix	focus, directrix	geometric definition
eccentricity	eccentricity	e (0=circle, 0<e<1=ellipse, e=1=parabola, e>1=hyperbola)
parametric curve	parametric equations	$(x(t), y(t))$
shortest distance	distance formula	$d = \frac{\lvert ax_0 + by_0 + c \rvert}{\sqrt{a^2+b^2}}$